



Institute of Computing

IT221 Data Structures and Algorithms Final Project Requirements

General Instructions

The following outlines the requirements for the group final project in IT221 Data Structures and Algorithms:

- Design and develop a software application that practically applies data structure and algorithmic concepts.
- Apply an algorithm related to data structures, binary trees, graph algorithms, or others that have not yet been discussed in the course.
- Provide a manuscript that discusses the theoretical and analytical aspects of your project.

System Development Instructions

- Propose a functional system that solves a specific problem or optimizes a particular process.
- The system must be developed using Java. You have the option to build either a:
 - Command-Line Interface (CLI) application.
 - Graphical User Interface (GUI) application.
- The proposed system must implement at least one advanced algorithm related to:
 - Data Structures
 - Binary Trees
 - Graph Algorithms
 - **Note:** *The chosen algorithm should be central to the system's core functionality, not just a minor feature. The algorithm should not yet have been discussed in the course.*

Algorithm Assessment Manuscript Instructions

- I. **System Overview**
 - a. **System Purpose and Problem Statement**
- II. **Algorithm Assessment**
 - a. **Background of the Algorithm**
 - b. **Time Complexity**
 - c. **Algorithm Mechanics and Logical Flow**
 - d. **Comparative Analysis**
- III. **Discussion**

The manuscript must adhere to the following requirements:

- **Format:** A4 Size
- **Spacing:** 1.5 Spacing
- **Margin:** Normal
- **Font:** 12 pt., Arial
- **Headings:** Must follow IEEE-style heading levels
- Any figures, tables, and equations should follow the IEEE format.
- References and in-text citations must be in IEEE format.

Video Discussion Instructions

The presentation must be a video recording of the discussion and simulation of the algorithm.

- The video must include discussion of the algorithm, simulation of the system, and the assessment conducted and its results.
- It must show the proponent's faces as well as use the proponent's voices.
- It must be creative and engaging.

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Stewardship
Adaptability and Excellence
Integrity and Innovativeness
Love of God and Country





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- It must not exceed 30 minutes.
- It must be uploaded to YouTube or Facebook.

Rubric and Criteria

Criteria	20	15	10	5	Total Maximum Score
Algorithm Implementation and Efficiency	The chosen algorithm is implemented flawlessly, highly optimized, and is central to the system's core function.	The algorithm is implemented correctly but may have minor inefficiencies or could be better integrated into the overall system.	The algorithm is partially implemented, contains logical bugs, or is too simplistic for the scope of the problem.	Missing, non-functional, or completely incorrect algorithm implementation.	20
Technical Manuscript and Complexity Analysis	Time and space complexity (Big O) are analyzed accurately for all scenarios (best, average, worst). The justification for the chosen algorithm is highly analytical and well-defended.	Complexity analysis is mostly correct but lacks depth. The justification is valid but relies on surface-level explanations.	Complexity analysis is incorrect or missing major components. Justification is weak or illogical.	Manuscript is poorly written, unformatted, or missing entirely.	20
System Functionality and Usability	The Java application (CLI or GUI) is fully functional, crash-free, user-friendly, and effectively solves the proposed problem.	The system works mostly as intended but has minor bugs, clunky navigation, or unhandled exceptions.	The system frequently crashes or fails to execute its primary purpose. UI/CLI is difficult to use.	The system does not compile or run.	20
Code Quality and Object-Oriented Design	Code is exceptionally clean, well-commented, strictly adheres to	Code is readable and mostly follows OOP, but lacks sufficient	Code is messy, hard to follow, lacks comments, or relies heavily	Incomprehensible or completely unorganized code.	20



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	Object-Oriented Programming (OOP) principles, and uses proper naming conventions.	comments or has minor structural flaws.	on procedural programming rather than OOP.		
System Video Presentation	Exceptional project presentation, optimal demonstration of key features, and compelling communication of the project's value.	Effective project presentation, clear demonstration of key features, and good communication of the project's value.	Basic presentation with room for improvement, some clarity in demonstrating features, and moderate communication of the project's value.	Ineffective presentation, lack of clarity in demonstrating key project features, and poor communication of the project's value.	20
				TOTAL	100

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